

# Application: are unlearned capabilities still in the weights?

The elicit-vs-teach metrics on WMDP-unlearned Zephyr-7B

**Question.** The main results show that the metrics separate a capability that is already in the weights (fine-tuning *elicits* it) from one that is absent (fine-tuning *teaches* it). Unlearning claims to move a model from the first case to the second. If it did, relearning is teaching and every metric should read “teach”; if the capability is only hidden, relearning is elicitation and the metrics should read “elicit”.

**Setup in six lines.** The original model is Zephyr-7B-beta. Four public unlearned versions of it are tested: RMU, ELM, NPO, SimNPO. The capability is WMDP-bio, 4-option multiple choice, scored as the argmax over the four answer letters (chance 0.25). The items are split at random into halves A and B. *Relearning* fine-tunes on A as text facts (question and correct answer, no options; LoRA  $r=64$ ) and is read on the held-out half B, which it never sees. The *fine-tuning null* is the same recipe on MMLU facts, which cannot teach bio answers. There is no never-learned model, so the elicit reference is the *original* model (before training) and the *original’s own relearning* (after training); the teach reference is each metric’s own null, and the elicit/teach values of the main results are given alongside.

**Reading a cell.** Each cell is the measured value and the call it makes. After relearning (Tables 1 and 3): **elicit** (reads like the original: at least half of the original’s distance from the null), *partly* (above its own null, less than half), *teach* (at its own null). Before training (Table 2) the only question is whether the capability can be read in the unlearned model: **readable**, *partly*, or not read. That reading is one-sided: what a metric reads is there, but what it cannot read may still be there, hidden before or after the point the metric looks at (Section 4). — means there is nothing to measure (the model has no task behaviour for this metric). A metric is called only where this design measures both its null and the original’s value; the others are shown next to the original’s value without a call (Table 4).

## 1. Does it come back, and what brings it back?

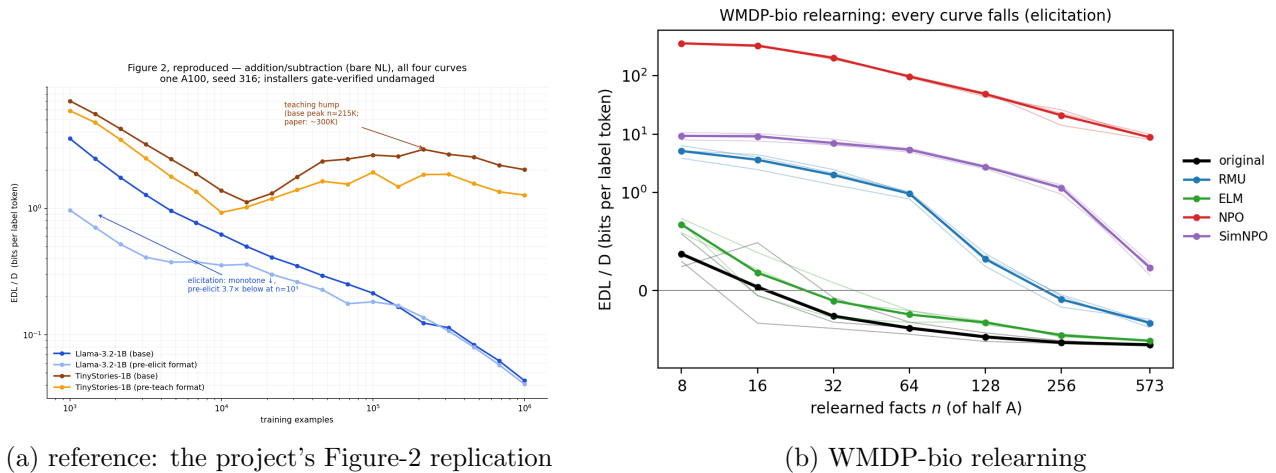
- **Every method leaves the capability in the weights:** relearning half the items brings the other half back to the original’s accuracy, and costs what the original’s own relearning costs.
- **What unlocks it differs:** for NPO and SimNPO any small fine-tune does; for RMU only bio-specific training does; ELM is in between.
- **The original paper’s own signature agrees** (M19, Figure 1): the information relearning extracts per label token falls with the number of facts for every method, in every seed and under both floors, as it does for the original. Teaching would show a rising phase, as the TinyStories arm does in panel (a). What an unlearned model needs beyond the original is paid on the first facts and shrinks with every doubling of  $n$ .

## 2. Before any training: is the capability still readable in the unlearned model?

- **SimNPO** reads like the original on 8 of 11: the circuit, the state, the heads and the probe are all intact. Only the output is pushed away from the answer.

|                                                         | original     | RMU               | ELM          | NPO         | SimNPO       |
|---------------------------------------------------------|--------------|-------------------|--------------|-------------|--------------|
| accuracy on B: the model as given                       | 0.67         | 0.31              | 0.35         | 0.26        | 0.42         |
| after an unrelated (MMLU) fine-tune                     | —            | 0.35              | 0.55         | 0.62        | 0.62         |
| after relearning on A                                   | —            | <b>0.68</b>       | <b>0.67</b>  | <b>0.64</b> | <b>0.66</b>  |
| M17 relearning beats the unrelated fine-tune            | —            | 0.68 vs 0.35      | 0.67 vs 0.55 | no room —   | no room —    |
| M17u the unrelated fine-tune unlocks B                  | —            | locked: 0.31→0.35 | 0.35→0.55    | 0.26→0.62   | 0.42→0.62    |
| M18 relearning cost over the original’s (nats)          | 0            | +0.15 ± 0.19      | +0.03 ± 0.14 | +4.6 ± 9.0  | −0.31 ± 0.53 |
| M19 EDL/D falls with $n$ (bits, $n=8 \rightarrow 573$ ) | 0.37 → −0.56 | 5.1 → −0.34       | 0.67 → −0.52 | 355 → 8.8   | 9.3 → 0.23   |
| MMLU (general ability)                                  | 0.58         | 0.58              | 0.59         | 0.47        | 0.48         |

Table 1: Accuracy on the held-out half B. M17: position between the unrelated-fine-tune value and the original, 90% interval RMU 1.04 [0.88, 1.19], ELM 0.98 [0.57, 1.40]; for NPO and SimNPO the unrelated fine-tune already reaches the original, so there is no room left to measure. M17u: position between the unlearned model and the original, RMU 0.11 [−0.02, 0.23], ELM 0.64 [0.50, 0.78], NPO 0.89 [0.78, 1.00], SimNPO 0.80 [0.62, 0.99]. M18: prequential code length of relearning (each batch scored before training on it) minus the original’s, net of the same difference on the unrelated fine-tune; 0 = as cheap as the original relearning facts it knows;  $\pm$  is 90%; NPO’s first steps repair a broken output, so its interval is too wide to read. M19: the original paper’s signature (Donoway et al., §4), EDL per label token of relearning on  $n$  facts of A, mean of three seeds, from  $n = 8$  to  $n = 573$ ; falling with  $n =$  elicitation, a rising phase = teaching (Figure 1).



(a) reference: the project’s Figure-2 replication

(b) WMDP-bio relearning

Figure 1: The EDL sweep. (a) The project’s replication of the paper’s Figure 2 (arithmetic; decisions.md 2026-08-22): the model that has the capability (Llama-3.2-1B, blue) falls monotonically; the model that must be taught it (TinyStories-1B, brown) falls, then rises 2.6 $\times$  (the teaching hump), then saturates. (b) Relearning on the first  $n$  facts of a seeded order of A, bits per label token, floor = the relearned model’s own validation loss (OCV; the paper’s test floor, the loss on B, gives the same calls); thick = mean of three seeds, thin = each seed; linear between  $-1$  and  $1$ , logarithmic beyond. Every curve falls, the original’s included; no method has a rising phase. The excess over the original per label token shrinks from  $n = 8$  to  $573$  by 21 $\times$  (RMU, 4.8  $\rightarrow$  0.22 bits), 7 $\times$  (ELM, 0.30  $\rightarrow$  0.04), 38 $\times$  (NPO, 355  $\rightarrow$  9.4) and 11 $\times$  (SimNPO, 9.0  $\rightarrow$  0.79): a fixed cost paid on the first facts, not a cost per fact. NPO’s level is its output repair (its first batch costs about 240 nats per label token). The original ends below zero because its floor, the validation rows (the first 10% of A in source order), is harder than the facts it trains on; one of its three seeds rises from  $n = 8$  to 16.

|                                             | metric                                                 | main<br>results:<br>elicit /<br>teach | original | RMU                               | ELM                       | NPO                  | SimNPO                   |
|---------------------------------------------|--------------------------------------------------------|---------------------------------------|----------|-----------------------------------|---------------------------|----------------------|--------------------------|
| M12                                         | hidden preference, bio (nats)                          | 2.31 / 0.06                           | 3.33     | 0.84 <i>partly</i>                | 0.40 <i>partly</i>        | 0.42 <i>partly</i>   | 3.52 <b>readable</b>     |
| M12                                         | hidden preference, cyber (nats)                        | 2.31 / 0.06                           | 3.57     | 0.34 <i>partly</i>                | 0.13 <i>partly</i>        | 2.32 <b>readable</b> | 3.31 <b>readable</b>     |
| M16'                                        | answer probe, bio (accuracy; shuffled $\approx 0.28$ ) | —                                     | 0.60     | 0.28 not read                     | 0.28 not read             | 0.47 <b>readable</b> | 0.49 <b>readable</b>     |
| M16'                                        | answer probe, cyber                                    | —                                     | 0.46     | 0.28 not read                     | 0.29 not read             | 0.41 <b>readable</b> | 0.39 <b>readable</b>     |
| M9                                          | answer depth: lens, best middle layer (nats)           | 2.1 / 0.0                             | 4.88     | 1.18 <i>partly</i>                | 0.39 <i>partly</i>        | 0.17 <i>partly</i>   | 0.45 <i>partly</i>       |
| M11                                         | repeatable task circuit (split-half J@32)              | 0.60 / noise                          | 0.64     | noise not read                    | noise not read            | noise not read       | 0.46 <b>readable</b>     |
| M14                                         | answer carried by the state (swap gain)                | +0.39 / +0.05                         | +0.66    | +0.08 not read                    | +0.09 not read            | +0.03 not read       | +0.38 <b>readable</b>    |
| M15                                         | option-reading heads (heads; flip rate)                | 22 / 0 heads                          | 15; 0.73 | 20; 0.52 <b>readable</b>          | 0 not read                | 0 not read           | 57; 0.70 <b>readable</b> |
| M1★                                         | original's 32 heads destroy U's answer (random 0.05)   | —                                     | 0.93     | 0.99 <b>readable</b> <sup>†</sup> | no answer —               | no answer —          | 0.89 <b>readable</b>     |
| M10★                                        | original's head states patched into U (accuracy)       | —                                     | 0.49     | 0.20→0.30 <i>partly</i>           | 0.27→0.48 <b>readable</b> | 0.00→0.00 not read   | 0.13→0.25 <i>partly</i>  |
| M2★                                         | wiring, U vs original (J@256; ceiling 0.55)            | 0.36 / —                              | —        | no answer —                       | no answer —               | no answer —          | 0.23 <i>partly</i>       |
| M13                                         | accelerating descent (negative share; blocks 20, 28)   | 1.00 / 0.00                           | 0.00     | 0.00 —                            | 0.00 —                    | <b>1.00</b> —        | 0.00 —                   |
| <b>tally</b> (readable / partly / not read) |                                                        |                                       |          | 2 / 4 / 4                         | 1 / 3 / 5                 | 3 / 2 / 4            | <b>8 / 3 / 0</b>         |

Table 2: Values on the unlearned model itself, no training. “Main results” gives the elicit parent and the teach parent from the arithmetic study where the metric has one. M16' adapts M16: the probe reads the correct option (4 classes), not the operands. M14: how often swapping the state at the answer position makes the model prefer the other item's answer, minus the rate with no swap. M10★: accuracy of U before and after the original's states at its 32 top heads are written in (32 random heads: RMU 0.20, ELM 0.34, SimNPO 0.16). <sup>†</sup> RMU's remaining answer signal is small (logit difference +0.75 vs +4.87 in the original); the original's heads carry all of it. M13 cannot be called here: the original is at a minimum of its own bio loss (negative share 0.00, unlike the main results' latent elicit parent at 1.00), so the metric has no elicit reference; NPO alone sits on a strongly concave slope ( $\lambda_{\min} = -1.5 \times 10^5$ ), where gradient ascent left it, and any descent step speeds up. Curvature is restricted to decoder blocks 20 and 28 (the full Hessian does not fit a 7B model on one GPU). M8 (latent reach) has no WMDP analogue and is not run.

- **NPO**: the answer is readable in the state (probe 0.47 vs original 0.60) while the output says nothing, so the damage is at the read-out.
- **ELM**: almost nothing is readable, but writing the original’s head states back in turns it on (0.27 → 0.48, the original scores 0.49 on this test).
- **RMU** hides it best: the probe, the circuit and the state swap cannot read it. Yet relearning brings it back in full (Table 1), so what these metrics miss in RMU is present, not absent.

### 3. After relearning on A: does the relearning look like eliciting or teaching?

Every relearned child is compared with the *original’s* relearned child, trained with the same recipe on the same facts the original already knows; that child is the elicit reference in this setting.

|                                                                                  | metric                                                       | main results:<br>elicit / teach | orig. re-learned | RMU                       | ELM                       | NPO                       | SimNPO                    |
|----------------------------------------------------------------------------------|--------------------------------------------------------------|---------------------------------|------------------|---------------------------|---------------------------|---------------------------|---------------------------|
| M1                                                                               | circuit overlap with the original, heads (J@32; chance 0.02) | 0.33 / 0.02                     | ceiling 0.60     | 0.73 <b>elicit</b>        | 0.64 <b>elicit</b>        | 0.68 <b>elicit</b>        | 0.68 <b>elicit</b>        |
| M1                                                                               | original’s 32 heads destroy the child’s answer (random 0.05) | 0.73 / –                        | 0.86             | 0.88 <b>elicit</b>        | 0.86 <b>elicit</b>        | 0.86 <b>elicit</b>        | 0.83 <b>elicit</b>        |
| M2                                                                               | wiring overlap with the original (J@256; chance 0.004)       | 0.36 / –                        | ceiling 0.27     | 0.32 <b>elicit</b>        | 0.32 <b>elicit</b>        | 0.33 <b>elicit</b>        | 0.29 <b>elicit</b>        |
| M4                                                                               | circuit at step 1 vs final (J@32; ceiling)                   | at step 1 / mid-run switches    | –                | 0.45 (0.78) <i>partly</i> | 0.73 (0.73) <b>elicit</b> | 0.49 (0.73) <b>elicit</b> | 0.42 (0.73) <i>partly</i> |
| M10                                                                              | child’s head states patched into U (accuracy)                | on / –                          | –                | 0.21→0.31 <i>teach</i>    | 0.33→0.54 <b>elicit</b>   | 0.00→0.00 <i>teach</i>    | 0.14→0.20 <i>teach</i>    |
| <b>tally, with Table 1 (elicit / partly / teach)</b>                             |                                                              |                                 |                  | 6 / 1 / 1                 | 8 / 0 / 0                 | 5 / 0 / 1                 | 5 / 1 / 1                 |
| <i>reported, not called (no null or teach value is measured for these here):</i> |                                                              |                                 |                  |                           |                           |                           |                           |
| M3                                                                               | head roles shared with the original (J; chance 0.01)         | 0.69 / 0.88                     | –                | 0.13                      | 0.08                      | 0.17                      | 0.15                      |
| M5                                                                               | gradient fades (first 1% / last 10%)                         | 3.4× / 0.7×                     | 3.2×             | 14.5×                     | 6.1×                      | 134×                      | 13.7×                     |
| M6                                                                               | weight write $\ \Delta W\ /\ W\ $ ( $\times 10^{-3}$ )       | 50 / 86                         | 8.8              | 8.9                       | 8.8                       | 11.2                      | 11.0                      |
| M7                                                                               | state change: one shared push (PC1)                          | 0.06 / 0.57                     | 0.64             | 0.49                      | 0.57                      | 0.69                      | 0.66                      |

Table 3: Values on the relearned child, read on B. “Original relearned” is the elicit reference: the same training on the original, which knows the facts. M1 and M2 ceilings are the split-half overlap of the circuits themselves; overlaps above the ceiling mean the child’s circuit is as close to the original’s as the original’s is to itself. M4: overlap of the circuit after the first training step with the final one, in brackets the ceiling. M10: accuracy of U with the child’s states at the child’s 32 top heads written in; the child itself scores 0.54–0.57. M3, M5, M6 and M7 are shown next to the original’s relearning but not called, because the main results’ scale does not carry over and this design has no teach value to replace it: the original’s own relearning reads 0.64 on M7, past the main results’ teach value; LoRA and the same stopping rule fix the size of M6’s write; M5’s first steps are inflated by repairing the unlearned output; and M3 did not separate elicit from teach in the main results either (0.69 vs 0.88). The tally counts M17, M18 and M19 from Table 1 (M17u is not counted: it asks what unlocks the capability, not whether it is there).

- **After relearning, all four read elicit on almost every called metric:** the child uses the original’s circuit (heads overlap 0.64–0.73 against a ceiling of 0.60), the original’s heads are as necessary in it as in the original’s own relearned child (0.83–0.88 vs 0.86), and the wiring is the original’s.
- The exception is patching a child’s states into the unlearned model (M10), which fails where the unlearned model’s own output is broken (NPO, SimNPO) or scrambled (RMU).

## 4. Why the calls are mixed, and how each metric is judged

Before training, a call can only be one-sided. Unlearning is meant to stop the capability from showing, so a metric that looks where the suppression acts, or after it, finds nothing, while one that looks before it, or goes around it, still finds the capability. Mixed calls therefore say where each method hides the capability; they do not contradict each other. Whether it is still in the weights is answered by the tests that bring it back (M17, M17u, M18, M19, M10★), and those read elicit for all four.

- **Hidden at the read-out: NPO and SimNPO.** The answer is readable in the state (probe 0.47 and 0.49; original 0.60, shuffled labels 0.28) while the output is pushed away from it (accuracy on B 0.26 and 0.42; original 0.67). Both methods train the output probability of the forget-set answers down. NPO was trained on bio only, and its cyber answers still show at the output (hidden preference 2.32; original 3.57).
- **Hidden before the answer forms: RMU and ELM.** Even the probe cannot read the answer, at any layer (0.28, the shuffled level), so the suppression acts inside the network before the state carries the answer. Both edit early and middle layers: ELM is a LoRA adapter on layers 4–7, and RMU steers the activations of layer 7 (WMDP paper). Going around that edit brings the capability back: writing the original’s head states into ELM restores its accuracy to 0.48 (original 0.49), and relearning restores both.

## Caveats

- There is no teach anchor from this model: without a model that never learned WMDP-bio, “elicit” means “like the original”, and the teach values come from each metric’s own null and from the main results.
- The metrics before training are read on all bio items, those after relearning on half B; the first 512 items in source order score lower (original 0.51) than B (0.67), so the two blocks are not compared with each other.
- The circuit metrics (M1, M2, M4 after relearning; M15 and M1★ before) compare the machinery that answers a four-option question. Much of it, reading the options and writing the letter, is likely shared across subjects, so an elicit reading there shows that the original’s answering machinery is in use, not that the bio knowledge is. The knowledge-specific evidence is the probe (M16′) and the recovery tests (M17 to M19). Two checks are queued (stage 1 and 3 additions, 2026-09-30, pending): the same overlaps against the original’s MMLU circuit, whose value is the cross-subject share, and the circuits on the relearning surface itself (question and answer, no options), where option-reading heads play no part.
- One seed (three for M19) and one relearning recipe; the checkpoint with the lowest validation loss is used because held-out accuracy moves by  $\pm 0.1$  between evaluations.
- M19 covers  $n \leq 573$ , all of half A. In Figure 1a the teaching hump comes after an initial fall and at far larger  $n$ , so a hump beyond 573 facts is not excluded by the sweep alone; but the teach arm there ends far above the elicit arm (2.0 vs 0.04 bits at  $n = 10^6$ ), while here the unlearned curves converge on the original’s.

|                         | metric                                  | null, measured on the same model  | elicit reference                            | called    |
|-------------------------|-----------------------------------------|-----------------------------------|---------------------------------------------|-----------|
| <i>before training</i>  |                                         |                                   |                                             |           |
| M12                     | hidden preference                       | the options permuted              | original                                    | yes       |
| M16'                    | answer probe                            | probe on shuffled labels          | original                                    | yes       |
| M9                      | lens, middle layers                     | zero logit difference ( $\pm$ SE) | original                                    | yes       |
| M11                     | repeatable task circuit                 | type-matched chance               | original                                    | yes       |
| M14                     | answer carried by the state             | rate with no swap                 | original                                    | yes       |
| M15                     | option-reading heads                    | no heads found                    | original                                    | yes       |
| M1★                     | original's heads necessary in U         | 100 random head sets              | original on itself                          | yes       |
| M10★                    | original's head states patched in       | 32 random heads                   | original (the donor)                        | yes       |
| M2★                     | wiring, U vs original                   | chance                            | split-half ceiling                          | yes       |
| M13                     | accelerating descent                    | none                              | none: the original sits at a minimum        | no        |
| M7★                     | state change, original to U             | the parent's own states           | describes the unlearning                    | no        |
| <i>after relearning</i> |                                         |                                   |                                             |           |
| M1, M2, M4              | circuit overlap with the original       | type-matched chance               | split-half ceiling                          | yes       |
| M1                      | original's heads necessary in the child | 100 random head sets              | original's relearned child                  | yes       |
| M10                     | child's head states patched into U      | 32 random heads                   | the child (the donor)                       | yes       |
| M17                     | held-out recovery                       | the fine-tuning null              | original                                    | yes       |
| M17u                    | an unrelated fine-tune unlocks B        | U itself                          | original                                    | yes       |
| M18                     | relearning cost                         | the fine-tuning null              | original's relearning                       | one-sided |
| M19                     | EDL sweep                               | the paper's shape rule            | original's own curve                        | yes       |
| M3                      | head roles                              | chance                            | none measured here                          | no        |
| M5                      | gradient fades                          | none                              | original's relearning, $3.2\times$          | no        |
| M6                      | weight write                            | none                              | original's relearning, $8.8 \times 10^{-3}$ | no        |
| M7                      | one shared push                         | none                              | original's relearning, 0.64                 | no        |

Table 4: What a single value is compared with. A metric is called only when its null and the original's value are both measured here; the rest are reported beside the original's value, since their thresholds come from the main results (full fine-tuning of a 1B model), whose scale does not carry over to LoRA on a 7B model.

- M19's OCV floor (the owner's default) sits on the harder validation rows; the paper's test floor (loss on B) lifts every curve by 0.37–0.41 bits per label token at  $n = 573$  (the original ends at  $-0.15$  instead of  $-0.56$ ) and changes no call.